

CORRESPONDENCE



Bariatric Surgery

Obesity management strategies should cut fat, not muscle

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The field of obesity management has seen rapid advances in the past decade. Until recently, the only guideline-endorsed options for obesity management were lifestyle intervention and bariatric surgery. The emergence of pharmacological interventions, in particular glucagon-like peptide-1 (GLP-1) agonists, has radically changed the obesity management landscape [1]. In this perspective, we wish to highlight one clinically important aspect that is generally neglected in obesity management research: the importance of maintenance of muscle mass.

CURRENT PRACTICES IN OBESITY MANAGEMENT RESEARCH

In virtually all existing obesity management trials, the effectiveness of the intervention is evaluated only through assessment of absolute and relative losses of total body weight [1]. To some extent, this focus is appropriate because loss of body weight is generally desirable and can drastically reduce morbidity and mortality in individuals with obesity. However, in addition to the kilograms lost, it is important to pay attention to the origin of the lost kilograms. It needs to be taken into account whether losses of body weight are attributable to loss of fat mass alone, fat and muscle mass, or even muscle mass alone.

MUSCLE AND ITS FUNCTIONS

Sufficient muscle mass and strength allow for activities of daily living and societal participation and thus affect quality of life [2]. Moreover, sufficient muscle mass and strength are prerequisites for physical activity which in turn has myriad beneficial metabolic, functional and clinical effects. In addition to its mechanical functions, muscle also plays a direct role in protein and glucose metabolism. For instance, muscle has the ability to store glucose in the form of glycogen, to an extent similar to the liver [3]. Muscle thus acts as a dynamic buffer of glucose homeostasis. Loss of muscle is associated with concomitant chronic inflammation, increased risk of infection, and frailty [2]. These detrimental effects of loss of muscle are largely attributable to dysregulation of mitochondrial networks, as skeletal muscle health and mitochondrial health are strongly intertwined [4].

THE CLINICAL RELEVANCE OF MUSCLE IN OBESITY MANAGEMENT

Following from the various functions of muscle, low muscle mass is consistently associated with poor metabolic, clinical and functional outcomes. The clinical importance of low muscle mass is particularly clear in individuals with obesity living with the so-

called 'double burden' of sarcopenic obesity [5], but also holds clinical relevance when individuals lose weight and no longer have obesity. For example, in a fully adjusted model, having a low muscle mass is associated with an estimated 37% and 47% higher risk of incident diabetes in males and females with a body mass index below 30, respectively [6].

Generally, obesity management interventions are accompanied by loss of muscle mass [7, 8]. Exemplary is the rapid initial loss of muscle mass after bariatric surgery [7]. Loss of muscle mass has, to varying extents, also been described for e.g. GLP-1 analogues [8]. However, this loss of muscle mass receives little attention in the obesity management literature and recent trials [1].

RECOMMENDATION FOR RESEARCH: EVALUATE POTENTIAL MUSCLE-WASTING EFFECTS OF OBESITY MANAGEMENT

Future studies should carefully assess short-term and long-term effects of obesity management on muscle mass and muscle function. This has recently gained further relevance, because continued long-term treatment appears to be necessary to maintain initial weight reductions [9].

The beneficial effects of the anti-obesity interventions on cardiovascular events and mortality are reassuring for the overall population undergoing these treatments [1]. Yet, these beneficial effects may be different across subgroups of patients. For example, any loss of muscle may be particularly important for patient groups vulnerable to sarcopenia, such as the elderly [5]. Therefore, future studies should not only provide population-based estimates of muscle status, but also identify subgroups of patients in whom muscle mass may decrease more and patients in whom loss of muscle mass may have particularly detrimental effects such as frail patients with sarcopenic obesity prior to the obesity management treatment.

There are multiple modalities to assess muscle status assessment in research. MRI has replaced the CT scan as the gold standard for measurement of muscle mass, and the DXA scan is a good alternative [10]. Moreover, handgrip strength and peak torque (measuring the strength of knee flexors and extensors) are reliable ways to assess muscle strength, and can be readily applied in trials virtually without any additional research costs.

RECOMMENDATIONS FOR THE CLINIC: DO NOT JUST AIM TO REDUCE FAT MASS, BUT ALSO TO MAINTAIN MUSCLE MASS

In clinical practice, caregivers should consider strategies to maintain muscle status when applying surgical or pharmacological obesity management therapies. Exercise as a parallel therapy holds particular promise for multiple reasons. Not only does exercise improve muscle mass and function, but it also holds potential to reverse or enhance the impaired mitochondrial function observed in aging and chronic muscle disuse [11].

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Furthermore, a high dietary protein intake may help to preserve muscle during weight loss [12]. In the future, muscle-preserving strategies may also include pharmacological strategies, if the benefits of therapies such as bimagrumab will be confirmed [13].

In conclusion, muscle is critically important for metabolism, functional and clinical outcome, regardless of body weight. Therefore, obesity management strategies should cut fat, not muscle. We plea for appropriate attention to muscle status in the evaluation of obesity management effects in both research and clinical practice.

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AUTHOR CONTRIBUTIONS

DK conceptualized and wrote the first manuscript draft. DS helped with the manuscript writing and revisions. SJLB and APB helped with the conceptualization and provided supervision.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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